

Urban Wünsch

Assistant Professor, Chalmers University of Technology

Researcher working on organic compounds dissolved in water — disentangling the composition of complex environmental samples to understand the fate of Dissolved Organic Matter (DOM) in aquatic systems. As of late 2026: 34 publications (10 as lead), H-index 19, 2100+ citations.

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Education

Technical University of Denmark

PhD 2018

University of Applied Sciences Bremerhaven

M.Sc. Biotechnology 2014

University of Applied Sciences Bingen

B.Sc. Biotechnology 2011

Experience

Chalmers University of Technology, Architecture and Civil Engineering

Assistant Professor 2025 – Present

DTU Aqua, National Institute for Aquatic Resources

Assistant Professor 2024 – 2025

Postdoc 2020 – 2024

Chalmers University of Technology, Architecture and Civil Engineering

Postdoc 2018 – 2020

Technical University of Denmark, National Institute for Aquatic Resources

PhD student 2014 – 2017

Teaching

Blue Bioeconomy Innovation

2023 – 2024

Course responsible

M.Eng, ~40 students

Ocean science & technology

2015 – 2025

Teaching topics in master-level courses

Machine learning for spectroscopic data analysis

2018 – Present

Co-organizer of online course

For Master- and PhD-students

Introduction to Chemistry & Urban spaces and functions for civil engineers

2018 –
2020

Teaching, bachelor-level courses

Large classrooms

Supervision

Julia Iwan (PhD)

2025 – Present

Main supervisor

Frederik Lenk (B.Sc.)

2024 – 2024

Main supervisor

Special course

Maria Papadimitraki (Research Assistant)

2023 – 2023

Co-supervisor

University of Southern Denmark

Emilie Tage Andresen (B.Sc.)

2022 – 2022

Co-supervisor

DTU Chemistry (Denmark)

Linea Gry Ebbesen (B.Eng.)

2022 – 2022

Co-supervisor

DTU Environment (Denmark)

Laila Vinther & Cecilie Hjelm Hvas Hansen (M.Sc.)

2021 – 2021

Co-supervisor, voluntary special course

DTU Environment (Denmark)

Evelyn Namuga Kasule (B.Sc.)

2021 – 2021

Co-supervisor

DTU Environment (Denmark)

Signe Melbye Andersen (M.Sc.)

2020 – 2022

Co-supervisor

DTU Aqua (Denmark)

Anders Dalhoff Bruhn Jensen (PhD)

2019 – 2023

Co-supervisor

DTU Aqua (Denmark)

Xianyu Kong (PhD)

2019 – 2022

PhD committee member

Alfred Wegener Institute (Germany)

Nashita Moona (PhD)

2018 – 2021

Co-supervisor

Chalmers (Sweden)

Skills

Dissolved organic matter characterization

Excitation-Emission Matrix (EEM) spectroscopy

Absorbance spectroscopy

High Performance Size Exclusion Chromatography (HPSEC)

Fourier Transform Ion Cyclotron Resonance Mass Spectrometry (FT-ICR-MS)

Chemometrics

Principal Component Analysis (PCA)

Parallel Factor Analysis (PARAFAC & PARAFAC2)

N-way Partial Least Squares (N-PLS)

Advanced Coupled Matrix and Tensor Factorization (ACMTF)

Research grants

Dissolved Organic Matter Biogeochemistry - innovative methods and sensors 2025

Strategic funding, Chalmers thematic area Ocean

2025-2029, Principal Investigator

A new approach to laccase discovery from the sea 2023

Independent Research Fund Denmark

2.5 million DKK, named applicant

New eyes on old GAC 2022

FORMAS, Sweden

146,000 SEK, co-applicant

Oxidation Demand and Rate (ODaR) as a sensitive proxy of biostability in drinking water 2021

FORMAS, Sweden

26,000 SEK, co-applicant

Enhancing Biogeochemical Fingerprints of Natural Organic Matter with Data Fusion 2019

Åforsk Foundation

400,000 SEK, main applicant

An optical sensor for reducing biocide dependence in chemical manufacturing 2019

Vinnova, Swedish Governmental Agency for Innovation Systems

474,000 SEK, co-applicant

2-year postdoctoral scholarship 2019

Carl Tryggers Stiftelse

552,000 SEK, main author

Conferences

Photosensitivity of fluorescent dissolved organic matter across the arctic ocean 2023

Aquatic Sciences Meeting 2023, Mallorca, Spain

Combining Physical and Mathematical Chromatography to Decipher the Molecular Fingerprints of Dissolved Organic Matter

2021

20th Conference of the International Humic Substances Society (online)

Mathematical chromatography deciphers the molecular fingerprints of dissolved organic matter

2021

Aquatic Sciences Meeting 2021 (online)

Membership in Scientific Communities

Association for the Sciences of Limnology and Oceanography (ASLO)

Member since 2014

International Humic Substances Society (IHSS)

Member since 2018